Adaptive Control

Introduction

- Basic Ideas in Adaptive Control
  - estimate uncertain plant / controller parameters on-line, while using measured system signals
  - use estimated parameters in control input computation
- Adaptive controller is a dynamic system with on-line parameter estimation
  - inherently nonlinear
  - analysis and design rely on the Lyapunov Stability Theory

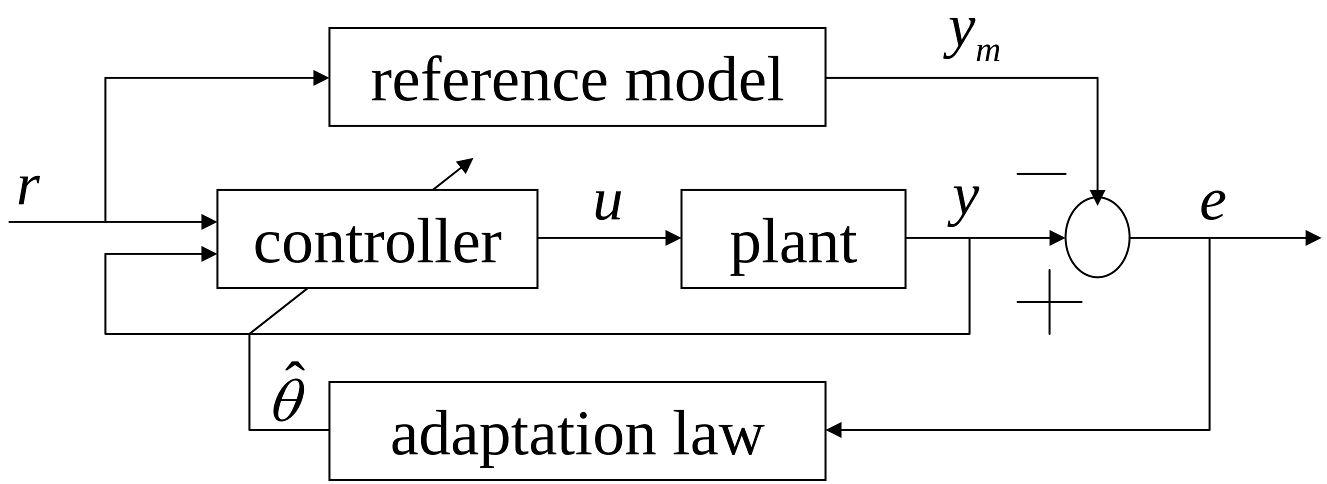
Historical Perspective

- Research in adaptive control started in the early 1950's
  - autopilot design for high-performance aircraft
- Interest diminished due to the crash of a test flight
  - X-?? aircraft tested
- Last decades witnessed the development of a coherent theory and many practical applications

Concepts

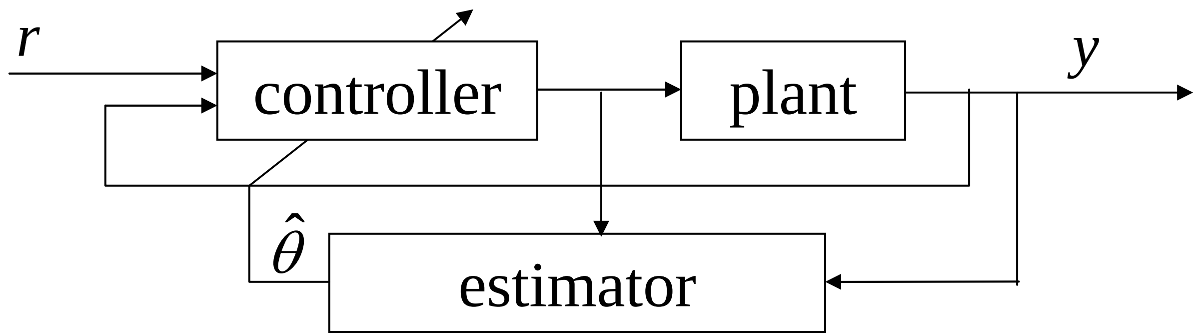
- **Why Adaptive Control?**
  - dealing with complex systems that have unpredictable parameter deviations and uncertainties
- **Basic Objective**
  - maintain consistent performance of a system in the presence of uncertainty and variations in plant parameters
- <u>Adaptive control</u> is superior to robust control in dealing with uncertainties in constant or slow-varying parameters
- <u>Robust control</u> has advantages in dealing with disturbances, quickly varying parameters, and unmodeled dynamics
- ***<u>Solution</u>***: Adaptive augmentation of a Robust Baseline controller

Model-Reference Adaptive Control (MRAC)



- *Plant* has a known structure but the parameters are unknown
- *Reference model* specifies the ideal (desired) response y_m to the external command r
- *Controller* is parameterized and provides tracking
- *Adaptation* is used to adjust parameters in the control law

Self-Tuning Controllers (STC)



- Combines a controller with an on-line (recursive) plant parameter estimator
- Reference model can be added
- Performs simultaneous parameter identification and control
- Uses <u>*Certainty Equivalence Principle*</u>
  - controller parameters are computed from the estimates of the plant parameters as if they were the true ones

Direct vs. Indirect Adaptive Control

- Indirect
  - estimate plant parameters
  - compute controller parameters
- Direct
  - no plant parameter estimation
  - estimate controller parameters (gains) only
- MRAC and STC can be designed using both Direct and Indirect approaches

MRAC Design of 1st Order Systems

- System Dynamics: $\boxed{\dot{x} = a\,x + b\left(u - f\left(x\right)\right)}$
  - $a\ ,\ b$ are constant ***unknown*** parameters
  - ***uncertain*** nonlinear function: $\boxed{f\left(x\right) = \sum_{i=1}^{N} \theta_i\, \varphi_i\left(x\right) = \theta^T\, \Phi\left(x\right)}$
    - vector of constant ***unknown*** parameters: $\theta = \left(\theta_1 \quad \ldots \quad \theta_N\right)^T$
    - vector of known basis functions: $\Phi\left(x\right) = \left(\varphi_1\left(x\right) \quad \ldots \quad \varphi_N\left(x\right)\right)^T$
- Stable Reference Model: $\boxed{\dot{x}_m = a_m\, x_m + b_m\, r, \quad \left(a_m < 0\right)}$
- ***Control Goal***
  - find u such that: $\boxed{\lim_{t\to\infty}\left(x\left(t\right) - x_m\left(t\right)\right) = 0}$

MRAC Design of 1st Order Systems (continued)

- Control Feedback: $\boxed{u = \hat{k}_x\, x + \hat{k}_r\, r + \hat{\theta}^T\, \Phi(x)}$
  - $(N + 2)$ parameters to estimate on-line: $\hat{k}_x, \hat{k}_r, \hat{\theta}$
- Closed-Loop System: $\boxed{\dot{x} = \left(a + b\,\hat{k}_x\right)x + b\left(\hat{k}_r\, r + \left(\hat{\theta} - \theta\right)^T \Phi(x)\right)}$
- Desired Dynamics: $\boxed{\dot{x}_m = a_m\, x_m + b_m\, r}$
- Matching Conditions Assumption
  - there exist *ideal* gains (k_x, k_r) such that: $\boxed{\begin{array}{l} a + b\,k_x = a_m \\ b\,k_r = b_m \end{array}}$
  - *Note*: knowledge of the ideal gains is not required, only their existence is needed
  - consequently:

$$\boxed{\begin{array}{l} a + b\,\hat{k}_x - a_m = a + b\,\hat{k}_x - a - b\,k_x = b\left(\hat{k}_x - k_x\right) = b\,\Delta k_x \\ b\,\hat{k}_r - b_m = b\,\hat{k}_r - b\,k_r = b\left(\hat{k}_r - k_r\right) = b\,\Delta k_r \end{array}}$$

MRAC Design of 1st Order Systems (continued)

- Tracking Error: $\boxed{e(t) = x(t) - x_m(t)}$
- Error Dynamics:

$$\boxed{\begin{aligned}
&\dot{e}(t) = \dot{x}(t) - \dot{x}_m(t) = \left(a + b\,\hat{k}_x\right)x + b\left(\hat{k}_r\, r + \underbrace{\left(\hat{\theta} - \theta\right)^T}_{\Delta\theta} \Phi(x)\right) - a_m\, x_m - b_m\, r \pm a_m\, x \\
&= a_m\left(x - x_m\right) + \left(a + b\,\hat{k}_x - a_m\right)x + b\left(\hat{k}_r - k_r\right)r + b\,\Delta\theta^T\,\Phi(x) \\
&= a_m\, e + b\left(\Delta k_x\, x + \Delta k_r\, r + \Delta\theta^T\,\Phi(x)\right)
\end{aligned}}$$

- Lyapunov Function Candidate:

$$\boxed{V\left(e(t), \Delta k_x(t), \Delta k_r(t), \Delta\theta(t)\right) = e^2 + |b|\left(\gamma_x^{-1}\,\Delta k_x^2 + \gamma_r^{-1}\,\Delta k_r^2 + \Delta\theta^T\,\Gamma_\theta^{-1}\,\Delta\theta\right)}$$

  – where: $\gamma_x > 0, \gamma_r > 0, \text{and } \Gamma = \Gamma^T > 0$ is symmetric positive definite matrix

MRAC Design of 1st Order Systems (continued)

- Time-derivative of the Lyapunov function

$$\dot{V}\left(e, \Delta k_x, \Delta k_r, \Delta\theta\right) = 2\,e\,\dot{e} + 2\left|b\right|\left(\gamma_x^{-1}\,\Delta k_x\,\dot{\hat{k}}_x + \gamma_r^{-1}\,\Delta k_r\,\dot{\hat{k}}_r + \Delta\theta^T\,\Gamma_\theta^{-1}\,\dot{\hat{\theta}}\right)$$

$$= 2\,e\left(a_m\,e + b\left(\Delta k_x\,x + \Delta k_r\,r\right) + \Delta\theta^T\,\Phi\left(x\right)\right)$$

$$+2\left|b\right|\left(\gamma_x^{-1}\,\Delta k_x\,\dot{\hat{k}}_x + \gamma_r^{-1}\,\Delta k_r\,\dot{\hat{k}}_r + \Delta\theta^T\,\Gamma_\theta^{-1}\,\dot{\hat{\theta}}\right)$$

$$= 2\,a_m\,e^2 + 2\left|b\right|\left(\Delta k_x\left(x\,e\,\mathrm{sgn}\left(b\right) + \gamma_x^{-1}\,\dot{\hat{k}}_x\right)\right)$$

$$+2\left|b\right|\left(\Delta k_r\left(r\,e\,\mathrm{sgn}\left(b\right) + \gamma_r^{-1}\,\dot{\hat{k}}_r\right)\right) + 2\,\left|b\right|\Delta\theta^T\left(\Phi\left(x\right)e\,\mathrm{sgn}\left(b\right) + \Gamma_\theta^{-1}\,\dot{\hat{\theta}}\right)$$

MRAC Design of 1st Order Systems (continued)

- Adaptive Control Design Idea
  - Choose adaptive laws, (on-line parameter updates) such that the time-derivative of the Lyapunov function decreases along the error dynamics trajectories

$$\boxed{\begin{aligned} \dot{\hat{k}}_x &= -\gamma_x \, x \, e \operatorname{sgn}(b) \\ \dot{\hat{k}}_r &= -\gamma_r \, r \, e \operatorname{sgn}(b) \\ \dot{\hat{\theta}} &= -\Gamma_\theta \, \Phi(x) \, e \operatorname{sgn}(b) \end{aligned}}$$

- Time-derivative of the Lyapunov function becomes semi-negative definite!

$$\boxed{\dot{V}\left(e(t), \Delta k_x(t), \Delta k_r(t), \Delta\theta(t)\right) = 2 \underbrace{a_m}_{<0} e(t)^2 \le 0}$$

MRAC Design of 1st Order Systems (continued)

- Closed-Loop System Stability Analysis
  - Since $\boxed{V \geq 0 \text{ and } \dot{V} \leq 0}$ then all the parameter estimation errors are bounded
  - Since the true (unknown) parameters are constant then all the estimated parameters are bounded
- Assumption
  - reference input $r(t)$ is bounded
- Consequently, x_m and $\dot{x}_m$ are bounded

MRAC Design of 1st Order Systems (continued)

- Since $x = e + x_m$ then x is bounded
- Consequently, the adaptive control feedback u is bounded
- Thus, $\dot{x}$ is bounded, and $\dot{e} = \dot{x} - \dot{x}_m$ is bounded, as well
- It immediately follows that $\boxed{\ddot{V} = 4a_m e(t)\dot{e}(t)}$ is bounded
- Using Barbalat's Lemma we conclude that $\dot{V}(t)$ is uniformly continuous function of time

MRAC Design of 1st Order Systems (completed)

- Using Lyapunov-like Lemma: $\boxed{\lim_{t\to\infty} \dot{V}(x,t) = 0}$
- Since $\boxed{\dot{V} = 2\,a_m\, e(t)^2}$ it follows that: $\boxed{\lim_{t\to\infty} e(t) = 0}$
- **Conclusions**
  - achieved asymptotic tracking: $\boxed{x(t) \to x_m(t), \text{ as } t \to \infty}$
  - all signals in the closed-loop system are bounded

Adaptive Dynamic Inversion (ADI) Control

ADI Design of 1st Order Systems

- System Dynamics: $\boxed{\dot{x} = a\,x + b\,u + f(x)}$
  - a, b are constant ***unknown*** parameters
  - ***uncertain*** nonlinear function: $\boxed{f(x) = \sum_{i=1}^{N} \theta_i\,\varphi_i(x) = \theta^T\,\Phi(x)}$
    - vector of constant ***unknown*** parameters: $\theta = (\theta_1 \quad \ldots \quad \theta_N)^T$
    - vector of known basis functions: $\Phi(x) = (\varphi_1(x) \quad \ldots \quad \varphi_N(x))^T$
- Stable Reference Model: $\boxed{\dot{x}_m = a_m\,x_m + b_m\,r, \quad (a_m < 0)}$
- **Control Goal**
  - find *u* such that: $\boxed{\lim_{t\to\infty}\left(x(t) - x_m(t)\right) = 0}$

ADI Design of 1st Order Systems (continued)

- Rewrite system dynamics:

$$\boxed{\dot{x} = \hat{a}\,x + \hat{b}\,u + \hat{f}(x) - \underbrace{(\hat{a} - a)}_{\Delta a}x - \underbrace{(\hat{b} - b)}_{\Delta b}u - \underbrace{\left(\hat{f}(x) - f(x)\right)}_{\Delta f(x)}}$$

- Function estimation error:

$$\boxed{\Delta f(x) \triangleq \hat{f}(x) - f(x) = \underbrace{\left(\hat{\theta} - \theta\right)}_{\Delta\theta}^{T} \Phi(x)}$$

- On-line estimated parameters: $\boxed{\hat{a},\quad \hat{b},\quad \hat{\theta}}$
- Parameter estimation errors

$$\boxed{\Delta a \triangleq \hat{a} - a,\quad \Delta b \triangleq \hat{b} - b,\quad \Delta\theta \triangleq \hat{\theta} - \theta}$$

ADI Design of 1st Order Systems (continued)

- ADI Control Feedback: $\boxed{u = \frac{1}{\hat{b}}\left(\left(a_m - \hat{a}\right)x + b_m\, r \;\; - \hat{\theta}^T\, \Phi\left(x\right)\right)}$
  - $(N + 2)$ parameters to estimate on-line: $\hat{a}, \hat{b}, \hat{\theta}$
  - Need to protect $\hat{b}$ from crossing zero
- Closed-Loop System: $\boxed{\dot{x} = a_m\, x + b_m\, r - \Delta a\, x - \Delta b\, u - \Delta\theta\, \Phi\left(x\right)}$
- Desired Dynamics: $\boxed{\dot{x}_m = a_m\, x_m + b_m\, r}$
- Tracking error: $\boxed{e \triangleq x - x_m}$
- Tracking error dynamics: $\boxed{\dot{e} = a_m\, e - \Delta a\, x - \Delta b\, u - \Delta\theta\, \Phi\left(x\right)}$
- Lyapunov function candidate

$$\boxed{V\left(e(t), \Delta a(t), \Delta b(t), \Delta\theta(t)\right) = e^2 + \gamma_a^{-1}\, \Delta a^2 + \gamma_b^{-1}\, \Delta b^2 + \Delta\theta^T\, \Gamma_\theta^{-1}\, \Delta\theta}$$

ADI Design of 1st Order Systems (continued)

- Time-derivative of the Lyapunov function

$$
\begin{aligned}
&\dot V\left(e,\Delta a,\Delta b,\Delta\theta\right)=2e\dot e+2\left(\gamma_a^{-1}\Delta a\,\dot{\hat a}+\gamma_b^{-1}\Delta b\,\dot{\hat b}+\Delta\theta^T\,\Gamma_\theta^{-1}\,\dot{\hat\theta}\right)\\
&=2e\left(a_m e-\Delta a\,x-\Delta b\,u-\Delta\theta\,\Phi\left(x\right)\right)\\
&+2\left(\gamma_a^{-1}\Delta a\,\dot{\hat a}+\gamma_b^{-1}\Delta b\,\dot{\hat b}+\Delta\theta^T\,\Gamma_\theta^{-1}\,\dot{\hat\theta}\right)\\
&=2a_m e^2+\Delta a\left(\gamma_a^{-1}\dot{\hat a}-x\,e\right)+\Delta b\left(\gamma_b^{-1}\dot{\hat b}-u\,e\right)+\Delta\theta^T\left(\Gamma_\theta^{-1}\dot{\hat\theta}-\Phi\left(x\right)e\right)
\end{aligned}
$$

- Adaptive laws

$$
\begin{aligned}
\dot{\hat a}&=\gamma_a\,x\,e\\
\dot{\hat b}&=\gamma_b\,u\,e\\
\dot{\hat\theta}&=\Gamma_\theta\,\Phi\left(x\right)e
\end{aligned}
\quad\Rightarrow\quad
\boxed{\dot V\left(e,\Delta a,\Delta b,\Delta\theta\right)=2a_m e^2\le 0}
$$

System energy decreases

ADI Design of 1st Order Systems (stability analysis)

- Similar to MRAC
- Using Barbalat's Lemma and Lyapunov-like Lemma: $\boxed{\lim_{t\to\infty}\dot{V}(x,t)=\lim_{t\to\infty}\left[2\,a_m\,e(t)^2\right]=0}$
- Consequently: $\boxed{\lim_{t\to\infty} e(t)=0}$ $\Rightarrow$ $\boxed{x(t)\to x_m(t),\ \text{as } t\to\infty}$
- **Conclusions**
  - asymptotic tracking
  - all signals in the closed-loop system are bounded

Parameter Convergence ?

- Convergence of adaptive (on-line estimated) parameters to their true unknown values depends on the reference signal $r(t)$
- If $r(t)$ is very simple, (zero or constant), it is possible to have non-ideal controller parameters that would drive the tracking error to zero
- Need conditions for parameter convergence

Persistency of Excitation (PE)

- Tracking error dynamics is a stable filter

$$\boxed{\dot{e}(t) = a_m\, e + b\underbrace{\left(\Delta k_x\, x + \Delta k_r\, r + \Delta\theta^T\, \Phi(x)\right)}_{\text{Input}}}$$

- Since the filter input signal is uniformly continuous and the tracking error asymptotically converges to zero, then when time t is large: $\boxed{\Delta k_x\, x + \Delta k_r\, r + \Delta\theta^T\, \Phi(x) \cong 0}$
- Using vector form:

$$\boxed{\begin{pmatrix} x & r & \Phi^T(x) \end{pmatrix} \begin{pmatrix} \Delta k_x \\ \Delta k_r \\ \Delta\theta \end{pmatrix} \cong 0}$$

Persistency of Excitation (PE) (completed)

- If $r(t)$ is such that $\boxed{v = \begin{pmatrix} x & r & \Phi^T(x) \end{pmatrix}^T}$ satisfies the so-called "*persistent excitation*" conditions, then the adaptive parameter convergence will take place
  - PE Condition: $\boxed{\exists \alpha > 0 \quad \forall t \quad \exists T > 0 \quad \int_t^{t+T} v(\tau) v^T(\tau)\, d\tau > \alpha I_{N+2}}$
- PE Condition implies that parameter errors converge to zero
  - for linear systems: m - sinusoids ensure convergence of ($2\,m$) - parameters
  - not known for nonlinear systems

ADI vs. MRAC

- No knowledge about $\operatorname{sgn} b$
- Adaptive laws are similar
- Both methods yield asymptotic tracking that does not rely on Persistency of Excitation (PE) conditions
- ADI needs protection against $\hat{b}$ crossing zero
  - If PE takes place and initial parameter $\hat{b}(0)$ has wrong sign then a control singularity may occur

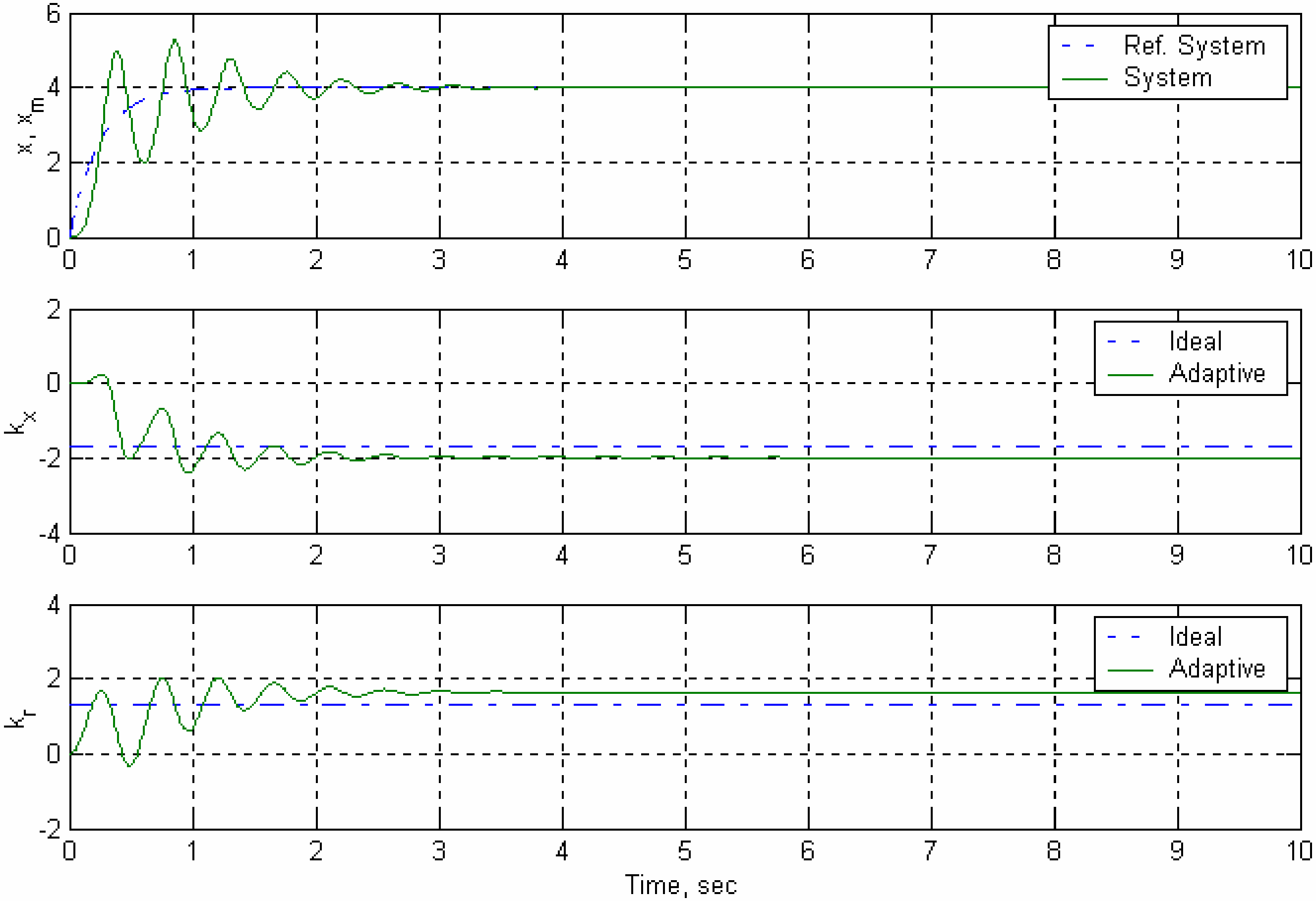
Example: MRAC of a 1st-Order *Linear* System

- Unstable Dynamics: $\boxed{\dot{x} = x + 3u, \quad x(0) = 0}$
  - plant parameters $\boxed{a = 1, \quad b = 3}$ are ***unknown*** to the adaptive controller
- Reference Model: $\boxed{\dot{x}_m = -4\,x_m + 4\,r(t), \quad x_m(0) = 0}$
- Adaptive Control: $\boxed{u = \hat{k}_x\, x + \hat{k}_r\, r}$

$$\boxed{\begin{aligned} a + b\,k_x &= a_m \\ b\,k_r &= b_m \end{aligned}}$$

- Parameter Adaptation: $\boxed{\begin{aligned} \dot{\hat{k}}_x &= -2\,x\,e, \quad \hat{k}_x(0) = 0 \\ \dot{\hat{k}}_r &= -2\,r\,e, \quad \hat{k}_r(0) = 0 \end{aligned}}$
- Two Reference Inputs: $\boxed{\begin{aligned} r(t) &= 4 \\ r(t) &= 4\sin(3t) \end{aligned}}$

1^{st}-Order *Linear* System
MRAC Simulation w/o PE: $r(t) = 4$

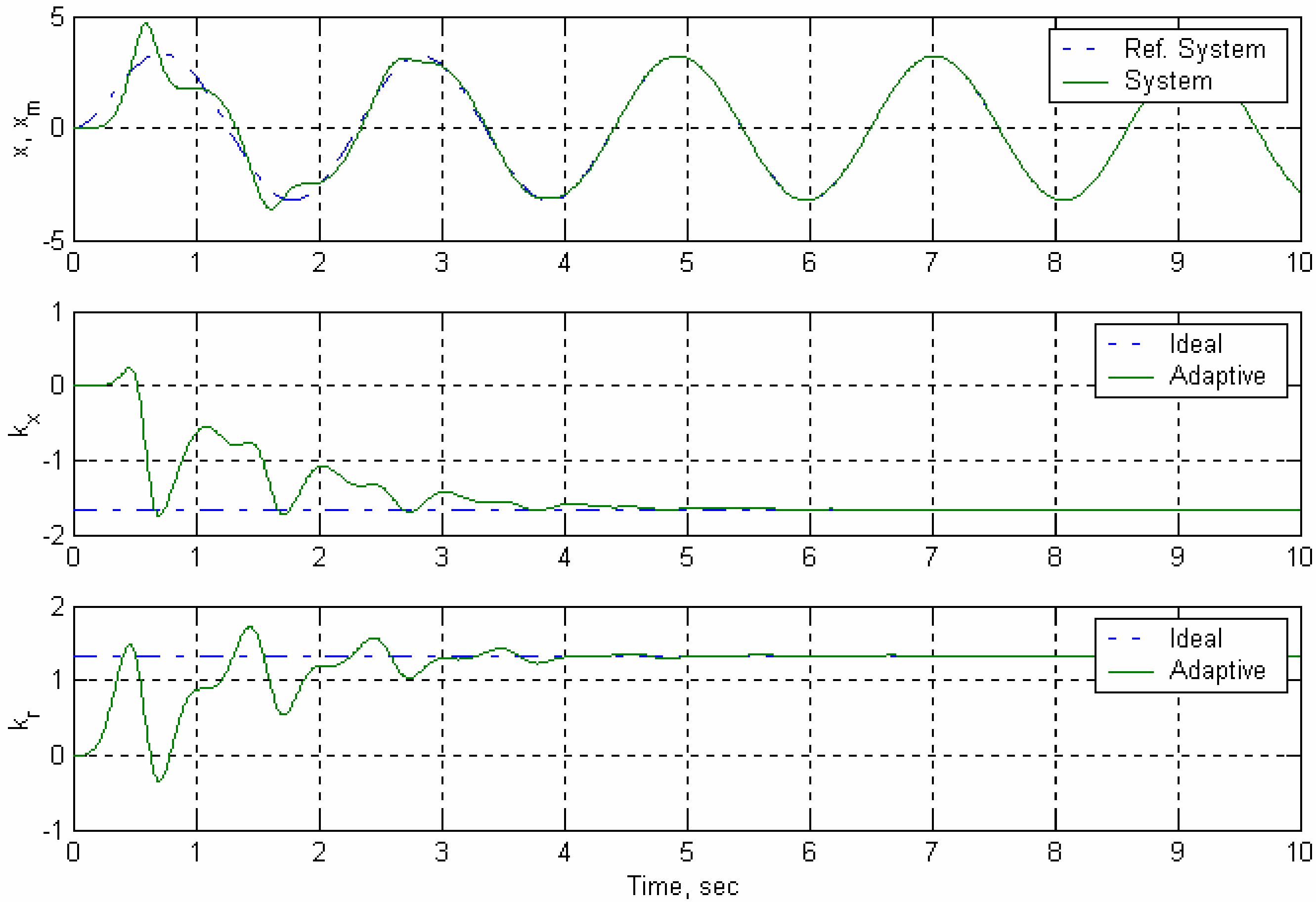


Tracking Error Converges to Zero

Parameter Errors don't Converge to Zero

1^{st}-Order *Linear* System

MRAC Simulation with PE: $r(t) = 4\sin(3t)$



Tracking and Parameter Errors Converge to Zero

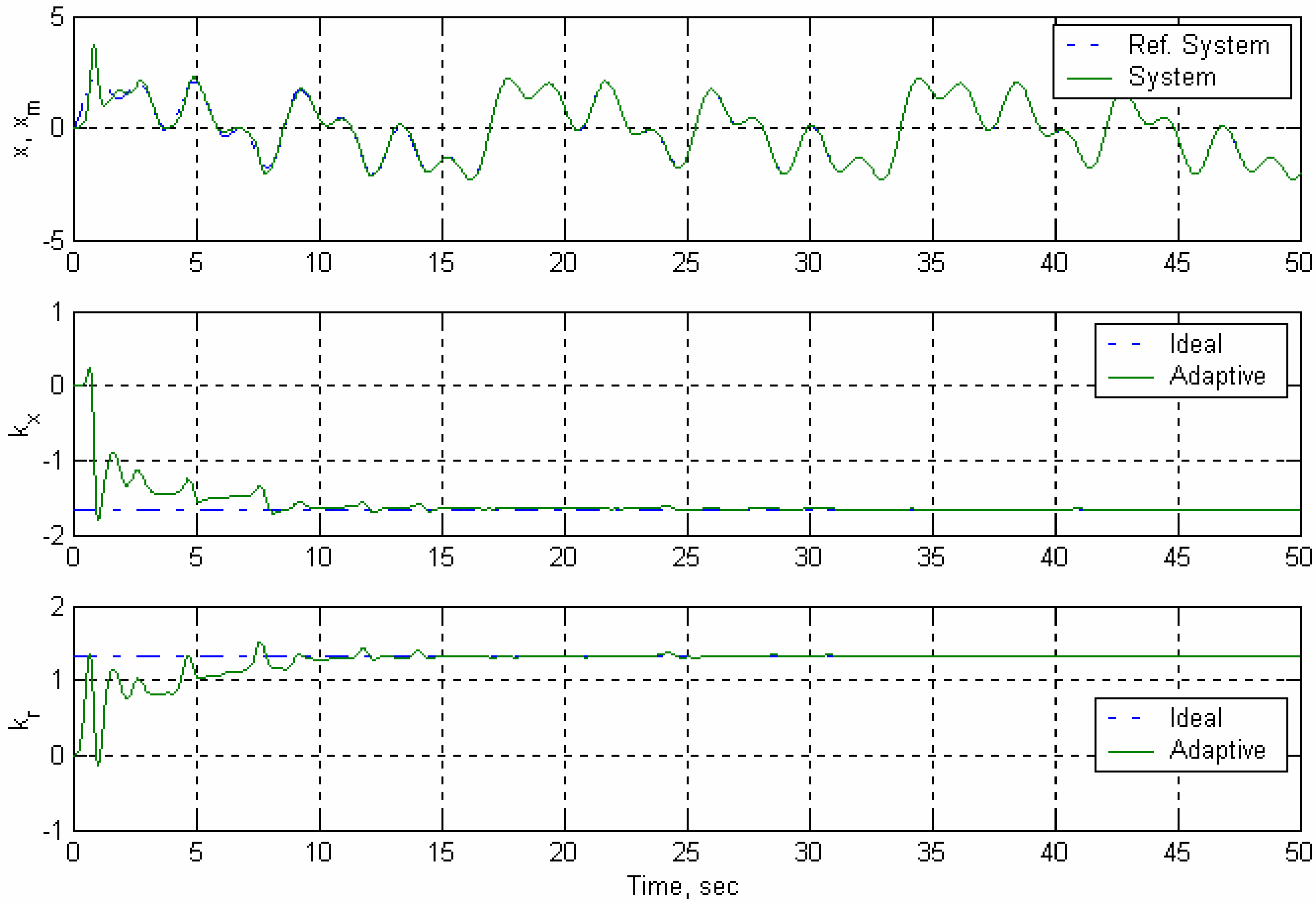
Example: MRAC of a 1st-Order *Nonlinear* System with *Local* Nonlinearity

- Unstable Dynamics: $\boxed{\dot{x} = x + 3\left(u - f(x)\right), \quad x(0) = 0}$
  - plant parameters $a = 1, \quad b = 3$ are *unknown*
  - nonlinearity: $f(x) = \theta^T \Phi(x)$
    - *known* basis functions: $\Phi(x) = \left(x^3 \quad e^{-(x+0.5)^2 10} \quad e^{-(x-0.5)^2 10} \quad \sin(2x) \right)^T$
    - *unknown* parameters: $\boxed{\theta = \left(0 \quad -1 \quad 1 \quad 0\right)^T}$
- Reference Model: $\boxed{\dot{x}_m = -4x_m + 4r(t), \quad x_m(0) = 0}$
- Adaptive Control: $\boxed{u = \hat{k}_x x + \hat{k}_r r + \hat{\theta}^T \Phi(x)}$
- Parameter Adaptation:

$$\boxed{\begin{aligned} \dot{\hat{k}}_x &= -2xe, \quad \hat{k}_x(0) = 0 \\ \dot{\hat{k}}_r &= -2re, \quad \hat{k}_r(0) = 0 \\ \dot{\hat{\theta}} &= -2\Phi(x)e, \quad \hat{\theta}(0) = 0_4 \end{aligned}}$$

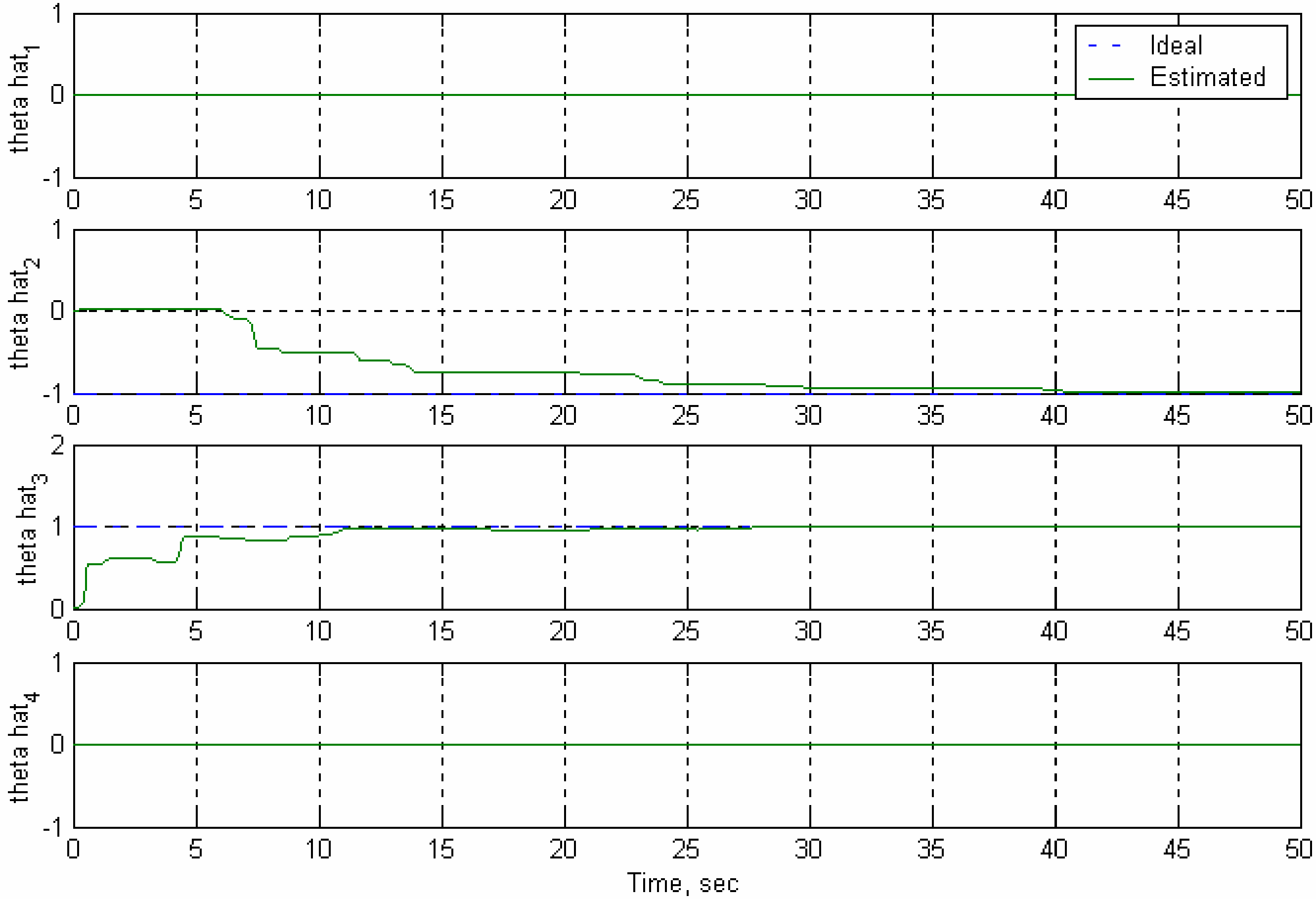
- Reference Input: $\boxed{r(t) = \sin(3t) + \sin\left(\frac{3t}{2}\right) + \sin\left(\frac{3t}{4}\right) + \sin\left(\frac{3t}{8}\right)}$

1^{st}-Order *Nonlinear* System with *Local* Nonlinearity: MRAC Simulation



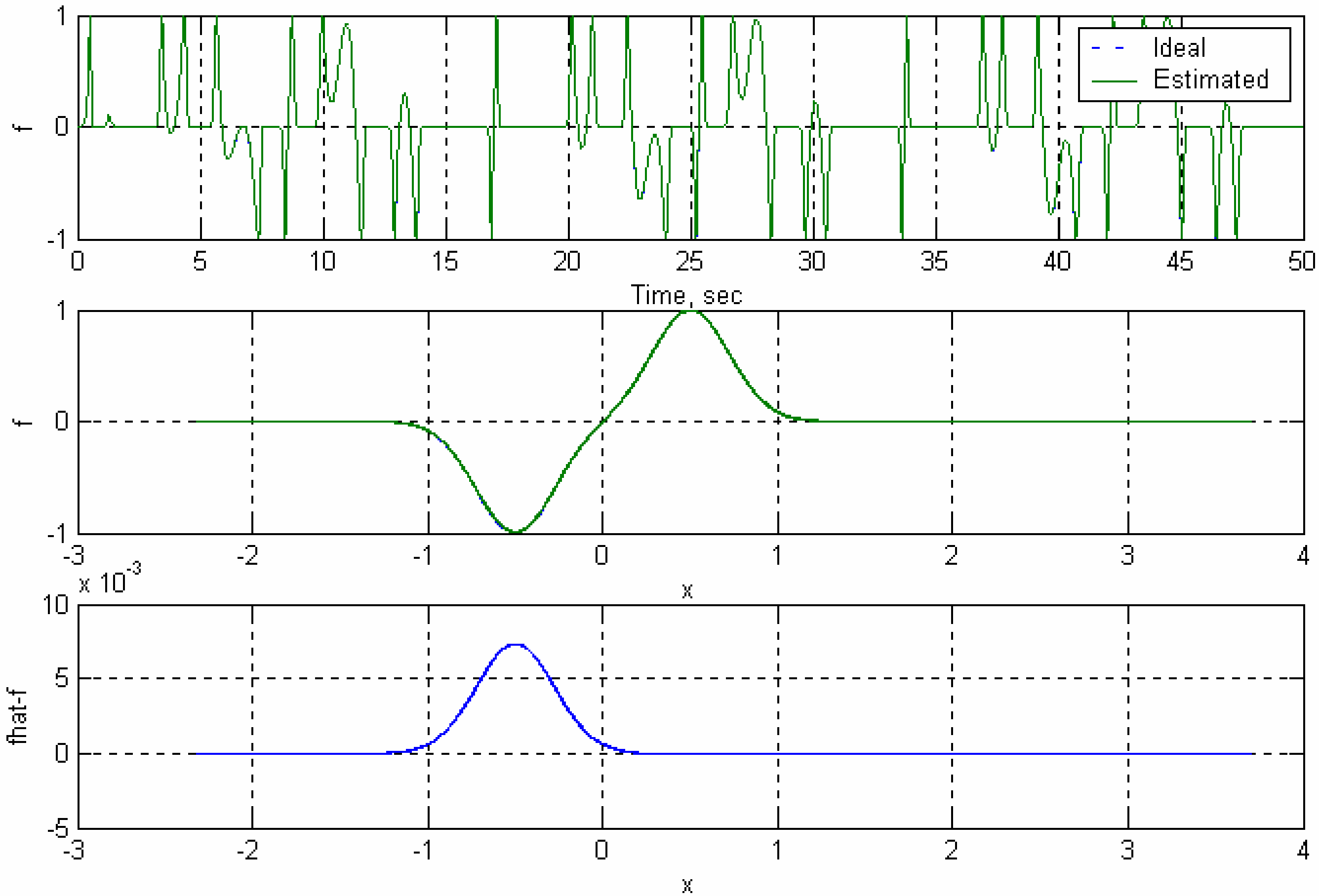
Good Tracking & Parameter Estimation

1^{st}-Order *Nonlinear* System with *Local* Nonlinearity: MRAC Simulation, (continued)



Nonlinearity: Good Parameter Estimation

1st-Order *Nonlinear* System with *Local* Nonlinearity: MRAC Simulation, (completed)



Nonlinearity: Good Function Approximation

MRAC of a 1st-Order *Nonlinear* System
Conclusions & Observations

- *Direct* MRAC provides good tracking in spite of unknown parameters and nonlinear uncertainties in the system dynamics
- Parameter convergence IS NOT guaranteed
- Sufficient Condition for Parameter Convergence
  - Reference input $r(t)$ satisfies Persistency of Excitation
    - PE is hard to verify / compute
  - Enforced for linear systems with *local* nonlinearities

MRAC Design of n^{th} Order Systems

- System Dynamics: $\boxed{\dot{x} = A\,x + B\,\Lambda\left(u - f(x)\right), \quad x \in R^n, \quad u \in R^m}$
  - $A \in R^{n\times n}, \Lambda = \mathrm{diag}\left(\lambda_1 \quad \ldots \quad \lambda_m\right) \in R^{m\times m}$ are constant ***unknown*** matrices
  - $B \in R^{n\times m}$ is ***known*** constant matrix
  - $\forall\, i = 1, \ldots, m \quad \mathrm{sgn}\left(\lambda_i\right)$ is ***known***
  - ***uncertain*** ***matched*** nonlinear function: $\boxed{f(x) = \Theta^T\,\Phi(x) \in R^m}$
    - ***matrix*** of constant ***unknown*** parameters: $\Theta \in R^{m\times N}$
    - vector of N ***known*** basis functions: $\Phi(x) = \left(\varphi_1(x) \quad \ldots \quad \varphi_N(x)\right)^T$
- Stable Reference Model: $\boxed{\dot{x}_m = A_m\,x_m + B_m\,r, \quad \left(A_m \text{ is Hurwitz}\right)}$

  $\boxed{r \in R^m, \quad A_m \in R^{n\times n}, \quad B_m \in R^{n\times m}}$
- **<u>Control Goal</u>**
  - find u such that: $\boxed{\lim_{t\to\infty}\left\|x(t) - x_m(t)\right\| = 0}$

MRAC Design of n^{th} Order Systems (continued)

- Control Feedback: $\boxed{u = \hat{K}_x^T\, x + \hat{K}_r^T\, r + \hat{\Theta}^T\, \Phi(x)}$
  - $(m\, n + m^2 + m\, N)$ - parameters to estimate: $\hat{K}_x,\quad \hat{K}_r,\quad \hat{\Theta}$
- Closed-Loop System: $\boxed{\dot{x} = \left(A + B\,\Lambda\,\hat{K}_x^T\right)x + B\Lambda\left(\hat{K}_r^T\, r + \left(\hat{\Theta} - \Theta\right)^T \Phi(x)\right)}$
- Desired Dynamics: $\boxed{\dot{x}_m = A_m\, x_m + B_m\, r}$
- Model Matching Conditions
  - there exist *ideal* gains (K_x, K_r) such that: $\Rightarrow$ $\boxed{\begin{aligned} A + B\,\Lambda\,K_x^T &= A_m \\ B\,\Lambda\,K_r^T &= B_m \end{aligned}}$
  - *Note*: knowledge of the ideal gains is not required

$$\boxed{\begin{aligned} A + B\,\Lambda\,\hat{K}_x^T - A_m &= A + B\,\Lambda\,\hat{K}_x^T - A - B\,\Lambda\,K_x^T = B\Lambda\left(\hat{K}_x - K_x\right)^T = B\,\Lambda\,\Delta K_x^T \\ B\,\Lambda\,\hat{K}_r^T - B_m &= B\,\Lambda\,\hat{K}_r^T - B\,\Lambda\,K_r^T = B\Lambda\left(\hat{K}_r - K_r\right)^T = B\,\Lambda\,\Delta\hat{K}_r^T \end{aligned}}$$

MRAC Design of n^{th} Order Systems (continued)

- Tracking Error: $\boxed{e(t) = x(t) - x_m(t)}$
- Error Dynamics:

$$
\begin{aligned}
\dot{e}(t) &= \dot{x}(t) - \dot{x}_m(t) = \\
&\left(A + B\Lambda \hat{K}_x^T\right)x + B\Lambda\left(\hat{K}_r^T r + \underbrace{\left(\hat{\Theta} - \Theta\right)}_{\Delta\Theta}{}^T \Phi(x)\right) - A_m x_m - B_m r \pm A_m x \\
&= A_m\left(x - x_m\right) + \left(A + B\Lambda \hat{K}_x^T - A_m\right)x + B\Lambda\left(\hat{K}_r - K_r\right)^T r + B\Lambda\Delta\Theta^T \Phi(x) \\
&= A_m e + B\Lambda\left(\Delta K_x^T x + \Delta K_r^T r + \Delta\Theta^T \Phi(x)\right)
\end{aligned}
$$

MRAC Design of n^{th} Order Systems (continued)

- Lyapunov Function Candidate

$$\boxed{\begin{aligned}&V\left(e,\Delta K_x,\Delta K_r,\Delta\Theta\right)=e^T P\, e\\&+\operatorname{trace}\left(\Delta K_x^T\,\Gamma_x^{-1}\,\Delta K_x\,|\Lambda|\right)+\operatorname{trace}\left(\Delta K_r^T\,\Gamma_r^{-1}\,\Delta K_r\,|\Lambda|\right)+\operatorname{trace}\left(\Delta\Theta^T\,\Gamma_\Theta^{-1}\,\Delta\Theta|\Lambda|\right)\end{aligned}}$$

  - where: $\operatorname{trace}(S)\triangleq\sum s_{ii}$
  - $|\Lambda|\triangleq\operatorname{diag}\left(|\lambda_1| \quad \ldots \quad |\lambda_m^i|\right)$ is diagonal matrix with positive elements
  - $\Gamma_x=\Gamma_x^T>0,\quad \Gamma_r=\Gamma_r^T>0,\quad \Gamma_\Theta=\Gamma_\Theta^T>0$ are symmetric positive definite matrices
  - $P=P^T>0$ is unique symmetric positive definite solution of the algebraic Lyapunov equation $\boxed{P\,A_m+A_m^T\,P=-Q}$
    - $Q=Q^T>0$ is any symmetric positive definite matrix

MRAC Design of n^{th} Order Systems (continued)

- Adaptive Control Design
  - Choose adaptive laws, (on-line parameter updates) such that the time-derivative of the Lyapunov function decreases along the error dynamics trajectories

$$\boxed{\begin{aligned}\dot{\hat{K}}_x &= -\Gamma_x \, x \, e^T P B \operatorname{sgn}(\Lambda) \\ \dot{\hat{K}}_r &= -\Gamma_r \, r \, e^T P B \operatorname{sgn}(\Lambda) \\ \dot{\hat{\Theta}} &= -\Gamma_\Theta \, \Phi(x) e^T P B \operatorname{sgn}(\Lambda)\end{aligned}}$$

- Time-derivative of the Lyapunov function becomes semi-negative definite!

$$\boxed{\dot{V}\left(e(t), \Delta K_x(t), \Delta K_r(t), \Delta\Theta(t)\right) = -e^T(t) Q e(t) \le 0}$$

MRAC Design of n^{th} Order Systems (completed)

- Using Barbalat's and Lyapunov-like Lemmas: $\boxed{\lim_{t\to\infty} \dot{V}(x,t) = 0}$
- Since $\boxed{\dot{V} = -e^T(t)\,Q\,e^T(t)}$ it follows that: $\boxed{\lim_{t\to\infty} \left\| e(t) \right\| = 0}$
- **Conclusions**
  - achieved asymptotic tracking: $\boxed{x(t) \to x_m(t), \text{ as } t \to \infty}$
  - all signals in the closed-loop system are bounded
- *Remark*
  - Parameter convergence IS NOT guaranteed

Robustness of Adaptive Control

- Adaptive controllers are designed to control real physical systems
  - non-parametric uncertainties may lead to performance degradation and / or instability
    - high-frequency unmodeled dynamics, (structural vibrations)
    - low-frequency unmodeled dynamics, (Coulomb friction)
    - measurement noise
    - computation round-off errors and sampling delays
- Need to enforce robustness of MRAC

Parameter Drift in MRAC

- When $r(t)$ is *persistently exciting* the system, both simulation and analysis indicate that MRAC systems are robust w.r.t non-parametric uncertainties
- When $r(t)$ IS NOT *persistently exciting* even small uncertainties may lead to severe problems
  - estimated parameters drift slowly as time goes on, and suddenly diverge sharply
  - reference input contains insufficient parameter information
  - adaptation has difficulty distinguishing parameter information from noise

Parameter Drift in MRAC: Summary

- Occurs when signals are not persistently exciting
- Mainly caused by measurement noise and disturbances
- Does not effect tracking accuracy until the instability occurs
- Leads to *sudden* failure

Dead-Zone Modification

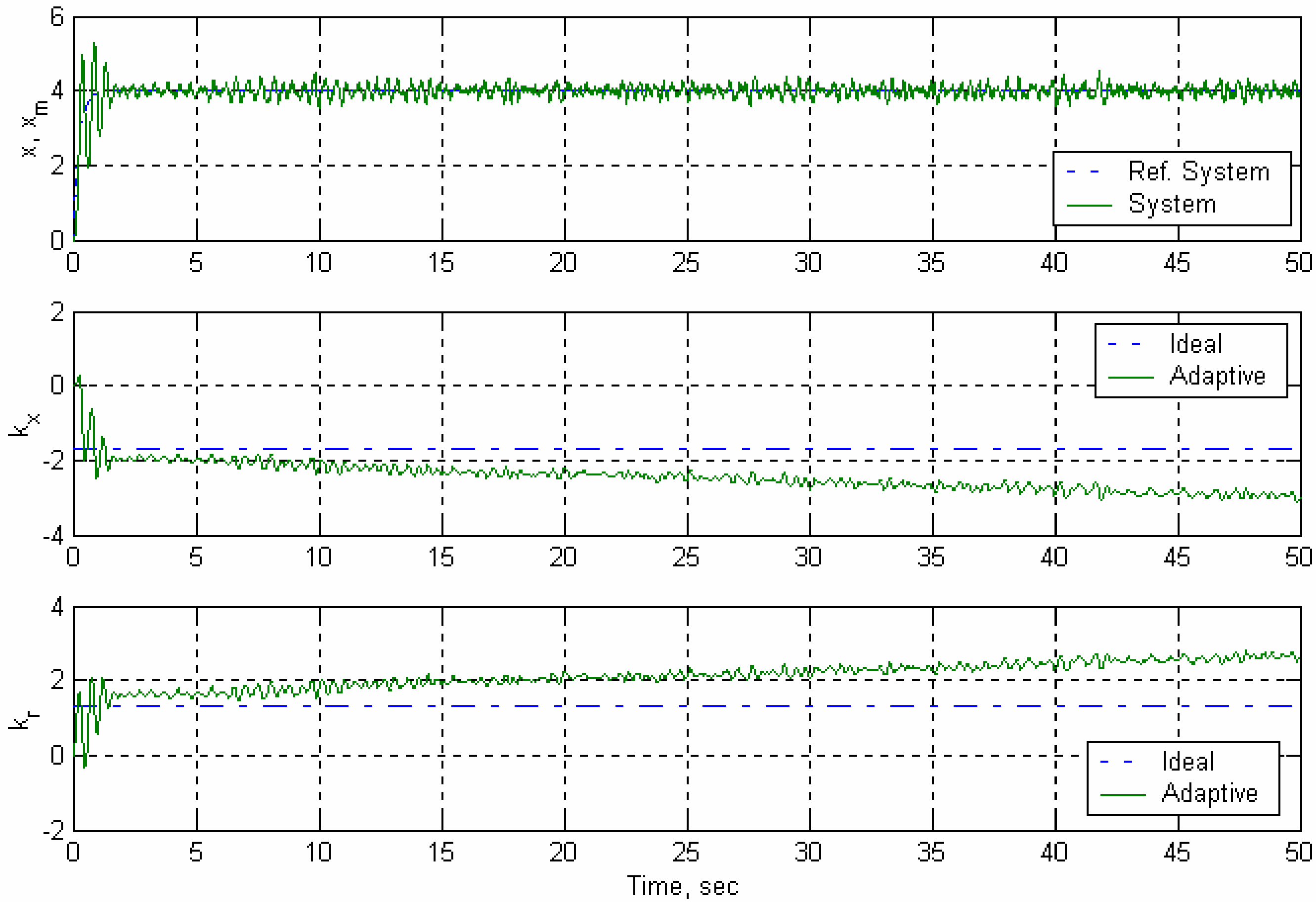
- Method is based on the observation that small tracking errors contain mostly noise and disturbance
- Solution
  - Turn off the adaptation process for “small” tracking errors
  - MRAC using Dead-Zone ⇨
  - ε is the size of the dead-zone
- Outcome
  - <u>*Bounded*</u> Tracking

$$\dot{\hat{K}}_x = \begin{cases} -\Gamma_x \, x \, e^T P B \operatorname{sgn}(\Lambda), & \|e\| > \varepsilon \\ 0, & \|e\| \le \varepsilon \end{cases}$$

$$\dot{\hat{K}}_r = \begin{cases} -\Gamma_r \, r \, e^T P B \operatorname{sgn}(\Lambda), & \|e\| > \varepsilon \\ 0, & \|e\| \le \varepsilon \end{cases}$$

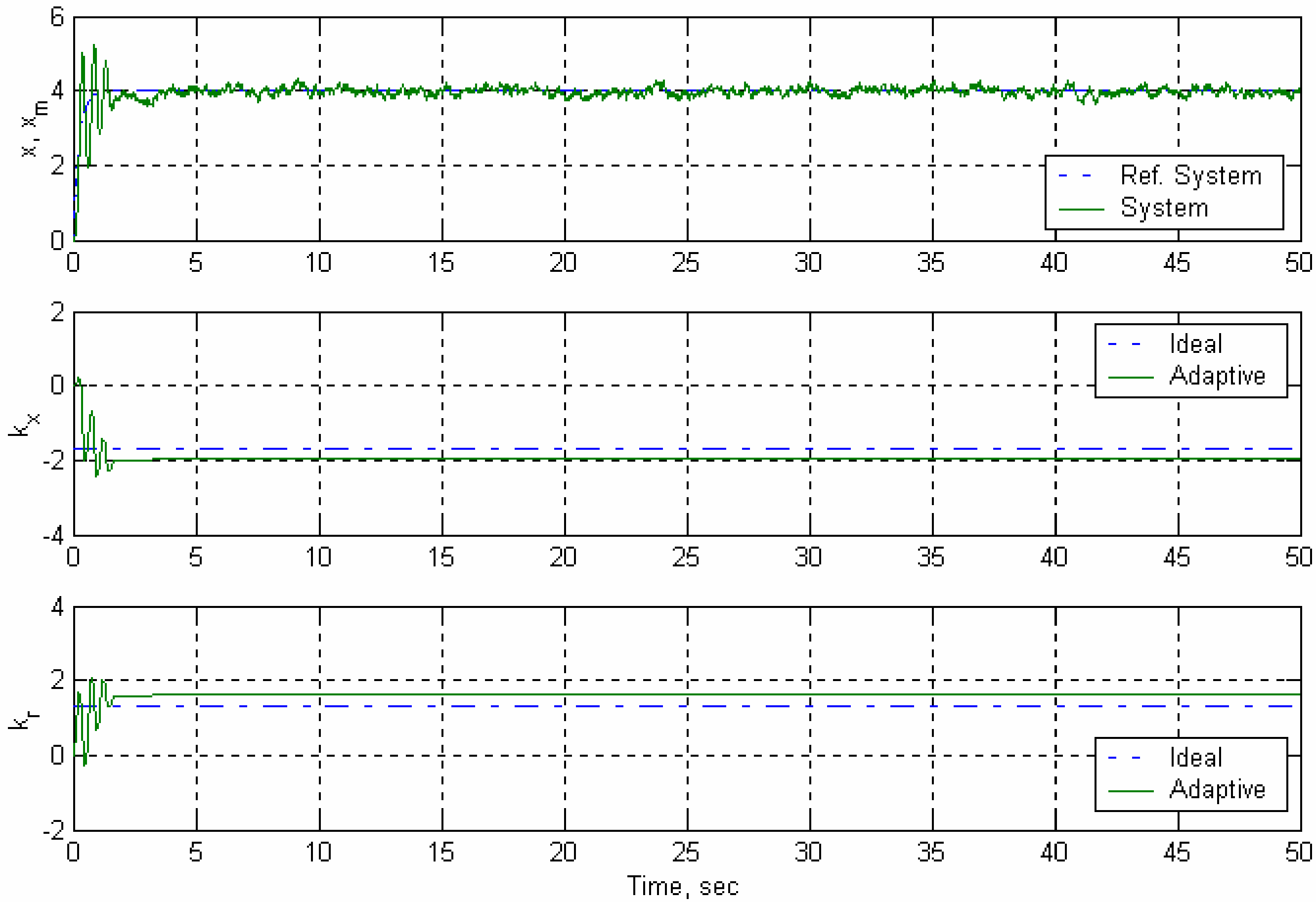
$$\dot{\hat{\Theta}} = \begin{cases} -\Gamma_\Theta \, \Phi(x) e^T P B \operatorname{sgn}(\Lambda), & \|e\| > \varepsilon \\ 0, & \|e\| \le \varepsilon \end{cases}$$

1^{st}-Order *Linear* System with Noise MRAC w/o Dead-Zone: $r(t) = 4$



- Satisfactory Tracking
- **Parameter Drift** due to measurement noise

1st-Order *Linear* System with Noise MRAC *with* Dead-Zone: $r(t) = 4$



- Satisfactory Tracking
- **No Parameter Drift**

Parametric and Non-Parametric Uncertainties

- Parametric Uncertainties are often easy to characterize
  - Example: $\boxed{m\ddot{x} = u}$
    - uncertainty in mass m is parametric
    - neglected motor dynamics, measurement noise, sensor dynamics are non-parametric uncertainties
- Both Parametric and Non-Parametric Uncertainties occur during Function Approximation

$$\boxed{\hat{f}(x) = \sum_{i=1}^{N} \underbrace{\theta_i}_{\text{parametric}} \varphi_i(x) + \underbrace{\varepsilon(x)}_{\text{non-parametric}}}$$

Enforcing Robustness in MRAC Systems

- Non-Parametric Uncertainty
  - Dead-Zone modification
  - Others ?
- Parametric Uncertainty
  - Need a set of basis functions that can approximate a large class of functions within a given tolerance
    - Fourier series
    - Splines
    - Polynomials
    - Artificial Neural Networks
      - sigmoidal
      - RBF